



Experimental study on the laminar flame speed of hydrogen/carbon monoxide/air mixtures

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ABSTRACT

Carbon monoxide and hydrogen are two important components in the syngas. In this study, the laminar flame speed of hydrogen/carbon monoxide fuel mixtures is measured over a large range of fuel compositions (0–100% volume fraction for hydrogen in the mixture) by using a Bunsen burner. The reaction zone area is used to calculate the laminar flame speed. The equivalence ratio covers from lean conditions to rich conditions. The experimental results show that by using the Bunsen flame, the laminar flame speed calculated with the reaction zone area is reliable. Based on the experimental results, empirical equations are derived which can be readily employed to calculate the laminar flame speeds of hydrogen, carbon monoxide, and hydrogen/carbon monoxide mixtures.

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1. Introduction

Coal is an important energy source in many countries, yet coal combustion could result in significant environment pollutions, such as acid rain and greenhouse effect, imposing serious global challenges. Therefore, the exploration of efficient and clean coal-based energy generation systems becomes a focal point of research. As one of the clean coal technologies with promising prospects, the polygeneration system presents an opportunity to overcome the aforementioned problems [1]. A polygeneration system, which can simultaneously produce electricity, chemical products, and clean synthetic fuels, may provide much higher efficiency and lower pollution than individual processes [2].

In the polygeneration system, the syngas is used as the fuel to generate electricity. Syngas fuels are primarily composed of H_2 and CO, and may contain N_2 , CO_2 , H_2O , CH_4 , and other higher-order hydrocarbons [3,4]. The specific compositions vary with the fuel source and processing technique. The variation of compositions and calorific value are the largest barriers to the usage of syngas. Therefore, it is of fundamental value to understand the impact of the variability on combustor performance and/or emissions, which requires an elucidation of the fundamental combustion properties of these mixtures.

The laminar flame speed is an important parameter of a combustible mixture, as it contains fundamental information of reactivity, diffusivity, and exothermicity. The value of the flame speed has important effects upon the propensity of the flame to

flashback and blowoff, and it also controls other key combustion characteristics, such as the flame's spatial distribution.

The flame speeds of the syngas-type mixture were measured in previous studies. For example, Scholte and Vaags [5] measured the burning velocities of mixtures of H_2 , CO, and CH_4 with Mach Hebra nozzle burners, however, only limited equivalence ratios were covered in their studies. Natarajan et al. [6] measured the laminar flame speeds of lean $H_2/CO/CO_2$ fuel mixtures by using the Bunsen burner and one-dimensional stagnation flame; the paper gave two components of H_2/CO ratios (50/50 and 5/95), and the equivalence ratio was emphasized on lean conditions only ($\phi = 0.6$ –1.05). Vagelopoulos and Egolfopoulos [8] measured the burning velocities of H_2/CO mixtures by using the counter flow flame technique. They reported that an addition of 6% or more hydrogen to H_2/CO could make the response of the mixture more akin to the kinetics of H_2 than that of CO. Brown et al. [9] reported the flame stretch effects on the burning velocities of pure H_2 -air, 50% H_2 -50% CO-air, and 5% H_2 -95% CO-air mixtures at the atmospheric condition. The values of Markstein length for 50% H_2 -50% CO-air mixtures were found to be very similar to those of pure H_2 -air mixtures. It was concluded that H_2 was the dominant specie and governed the Markstein length behavior for the 50% H_2 -50% CO-air mixture.

Most of these previous works investigated limited equivalence ratios and fuel compositions. For practical applications, it is necessary to extend the range of the flame speed data for syngas mixtures. In the current study, we measure the laminar flame speed of H_2/CO mixtures over a wide range of fuel compositions (0–100% for H_2 and CO by volume) and equivalence ratios, based on the flame area images of a conical Bunsen flame. Moreover,

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numerical simulations of the laminar flame speed on H_2/CO mixtures are carried out and compared with experimental results.

2. Bunsen flame method

The Bunsen flame is one of the oldest known examples of stabilized premixed combustion. Most importantly, it provides one of the simplest models of a multidimensional combustion field. Since the invention of the first laboratory premixed flame burner by Bunsen in 1855, many researchers have adopted this model burner for studying the premixed flames, which were concluded in the extensive review by Lewis and Von [10].

The laminar flame speed is defined as the velocity that a planar flame front travels relative to the unburned gas in a direction normal to the flame surface [11]. Although the laminar flame speed is unambiguous in definition, it is difficult to be measured exactly in practice. Hence some assumptions have to be made in practical measurements. A flame stabilized on the rim of a cylindrical burner is conical in shape and not one-dimensional. This conical flame is affected by its strain and curvature – the influence of both factors on the local flame speed also depends on the Markstein length of the mixture [11]. In this study, the flame speed measurement is based on a weighted average of the area over the entire flame surface. The average flame speed is calculated by dividing the volume flow rate of the mixture by the flame area. It is clear from the definition of the unburned laminar flame speed that the true flame area should be the unburned flame area, i.e. the upstream of the preheat zone of the flame, as shown in Fig. 1. However, as stated in Ref. [6], the flame speed calculated based on the reaction zone area may provide a result that matches more closely with the unstretched flame speed. Based on the mass balance of the flame front, the mass flow rate is

$$\dot{m} = S_b \rho_b A_b, \quad (1)$$

where S_b is the flame speed of the burned mixture, ρ_b is the density of the burned mixture and A_b is the surface area at the end of the heat release zone. According to the mass balance of the flame

$$S_b \rho_b = S_u \rho_u, \quad (2)$$

where S_u is the flame speed of unburned mixture. And from Eqs. (1) and (2), we can obtain

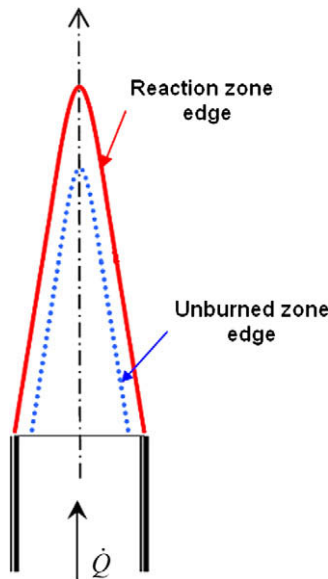


Fig. 1. Bunsen flame with two edges.

$$S_u = \frac{\dot{m}}{\rho_u A_b} = \frac{\dot{Q}}{A_b}, \quad (3)$$

where $\dot{Q}$ is the volumetric flow rate of the unburned mixture. In the present study, the flame speed is calculated by dividing the volumetric flow rate of the mixture by the luminous cone surface area.

3. Experimental setup and procedure

Fig. 2 is a schematic of the experiment setup used for laminar flame speed measurements. The experiment setup includes a Bunsen burner (premixed burner), gas supply system, high accuracy flux controlling and mixing systems, and system for measuring the flame speed.

The desired volume of air, CO, and H_2 are first measured by thermal mass flow meters (MFM for short), one for each gas phase. CO and H_2 are mixed in a tube which is long enough for uniform mixing. Next, the mixture of CO and H_2 enters the Bunsen burner. At the same time, the required quantity of air is added to Bunsen burner. The thermal MFM has the advantages of high accuracy, great repeatability, quick response speed, and stability. In particular, it can induce stabilizing action on the gas fluctuation. The thermal MFMs are calibrated with a standard procedure, by using N_2 of 0 °C and 1 atm. While measuring other gases, a corresponding conversion coefficient should be used to calculate the actual gas flow rate. The accuracy of MFM is $\pm 1.5\%$ within its full scale.

The design of Bunsen burner is of primary importance for the present research. Two burners with inner diameter (D) 4 and 5 mm are employed; each one is a straight copper tube. A copper net set is placed downstream of mixture in the tube, which can yield even mixture at the burner exit. The length of each tube is at least 50D to ensure that the flow is laminar and that the exit velocity profile is fully developed [6]. The burner diameter is chosen to ensure that the flow remains laminar (Reynolds number, $Re_D < 2000$) and the flame prevents flashback and blowoff.

Digital images of the flame are captured with a 12-bit scientific charge-coupled device camera (1600 × 1200 pixels) and a 18–108 mm, $f/2.5$ camera lens. The camera system is of high sensitivity, large dynamic range and large well capacity. The flame pictures can be captured clearly by the camera. Fig. 3 shows some typical images of the flames including lean and rich conditions, and low and high H_2 composition mixtures. With such information, one can distinguish the flame reaction zone easily.

During the experiment, the volumetric flow rates of the fuel and air are recorded separately. In order to eliminate the influence of small fluctuations to the flame speed, we capture 20 pictures of flame for each experimental condition.

Since the Bunsen flames are essentially axisymmetric, each flame image is split in half along the burner axis. For every half flame image, Simple PCI software is used to detect the edge of the luminous cone surface by locating the maximum derivative of the flame intensity along the radius of the flame. The flame area is then calculated by revolving the detected edge along the axis of the burner. The same procedure is repeated for the other half of the flame image. For each experimental condition, the reaction zone area A_b is determined by the average of the 40 area values. The laminar flame speed can be then calculated by Eq. (3).

4. Results and discussion

4.1. Laminar flame speed of H_2 /air and CO/air

Fig. 4 shows the laminar flame speed versus the equivalence ratios for H_2 , the results from previous works are also given. For H_2 /air mixtures, the burner diameter is chosen to be 4 mm. The laminar flame speeds of H_2 /air mixtures in previous works are slightly

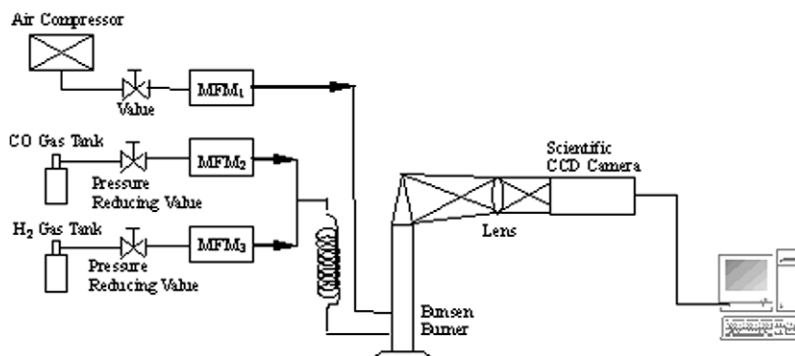


Fig. 2. Schematic of the experiment setup (MFM = Mass Flow Meter).

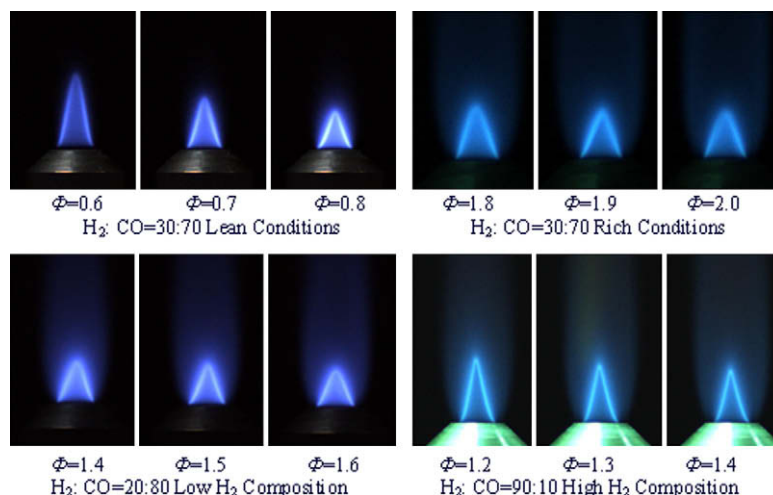


Fig. 3. Images of flame for various equivalence and composition.

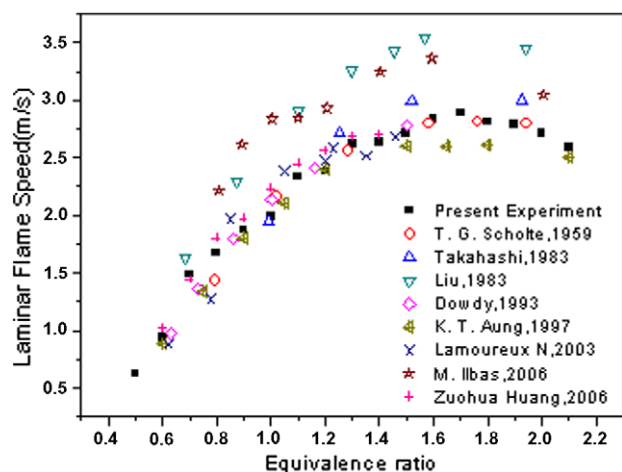


Fig. 4. Laminar flame speeds of H₂–air mixtures versus equivalence ratio (see [5,12–18]).

different. The laminar flame speeds reported in Liu and Macfarlane [12] and Lbas et al. [13] are larger than our work, but the trend of laminar flame speeds versus equivalence ratio is the same. Our work agrees well with works of Scholte and Vaags [5], Takahashi et al. [14], Dowdy et al. [15], Aung et al. [16], Lamoureux et al. [17], and Huang et al. [18].

Fig. 5 gives the laminar flame speed versus the equivalence ratios for CO. The burner diameter is chosen to be 5 mm. The data of

pure CO/air's laminar flame speeds are scarce in previous work (most previous works studied the H₂/CO/air mixtures with 1% H₂). Our measured laminar flame speeds agree well with Rightley's study [19], but are smaller than the works of Sun et al. [7] and Bouvet et al. [20]. The main difference is caused by the fact that in Refs. [7,20], there was 1% H₂ in H₂/CO mixture.

To briefly summarize, the good agreements between our work and the previous work indicate that the errors in our measurement

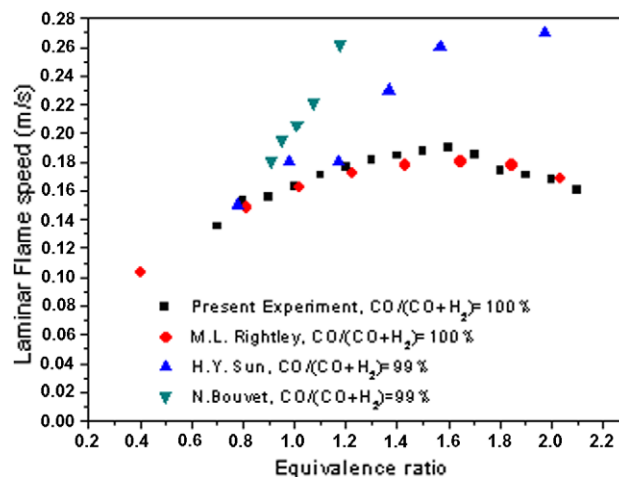


Fig. 5. Laminar flame speeds of CO/air mixtures versus equivalence ratio (see [19,7,20]).

of the flame speed (which is associated with reaction zone area based on Bunsen method) are small, and this technique is reasonably accurate. When the equivalence ratio equals to 1.7 and 1.6, the laminar flame speeds of H_2 /air mixtures and CO/air mixtures reach their maximum values 2.891 and 0.19 m/s, respectively.

4.2. Laminar flame speed of H_2 /CO mixtures with air: experimental results and numerical simulation

Laminar flame speeds of H_2 /CO with air versus the equivalence ratio are given in Fig. 6, as the hydrogen ratio varies from 10% to 70% at full equivalence ratios, and from 80% to 90% at lean conditions, the 5 mm tube is used. Only at high hydrogen levels (80% H_2 and 90% H_2) and at rich conditions, the 4 mm tube is employed to prevent flashback. For a given equivalence ratio, the laminar flame speed increases with H_2 fraction. The laminar flame speed of H_2 is much greater than that of CO and H_2 /CO mixtures. A big increase in S_L is observed when the H_2 fraction is increased to 10%. In order to investigate the influence of H_2 fraction on the laminar flame speeds of H_2 /CO mixtures more clearly, the laminar flame speed versus H_2 fraction is given in Fig. 7, when the equivalence ratio equals to 1.0. The curve can be divided to three parts. The first part is distinguishable when the H_2 fraction below 20%, where the laminar flame speed increases rapidly with H_2 fraction. This is mainly because the CO oxidation reaction is sensitive to the presence of small amounts of H_2 in mixture. The second part is distinguished when the H_2 fraction is among 20–80%, the laminar flame speed increases with H_2 fraction in a nearly linear trend. This is due to the fact that the sensitivity of the CO oxidation reaction to H_2 fraction is reduced. The third part refers to the section when the H_2 fraction is beyond 80%, where the characteristic of mixtures is more close to H_2 , and thus there is a higher sensitivity of the laminar flame speed to the H_2 fraction, and the laminar flame speed increases rapidly with H_2 fraction.

The unstrained 1d laminar flame speeds are calculated with the PREMIX algorithm of the Chemkin package. The Davis' mechanism [21], which was developed specifically for H_2 /CO combustion, is employed in the present numerical simulation. The Davis mechanism is an optimized kinetic model which consists of 14 species and 30 reactions, and it incorporates recent updates for rate parameters and third-body efficiencies of a few key reactions. In this study, the laminar flame speeds of 30% H_2 –70% CO and 50% H_2 –50% CO are calculated at the equivalence ratio 0.7–2.1, and the results are shown in Fig. 8. For comparison purposes, the experimental results are also given. The figure show that the Davis

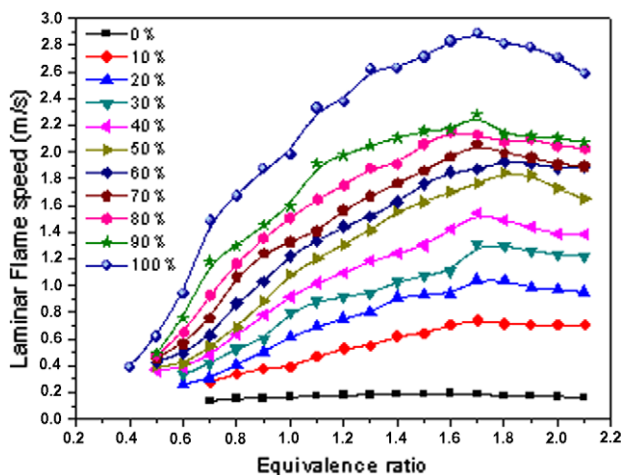


Fig. 6. Laminar flame speeds of H_2 /CO mixtures versus equivalence ratio.

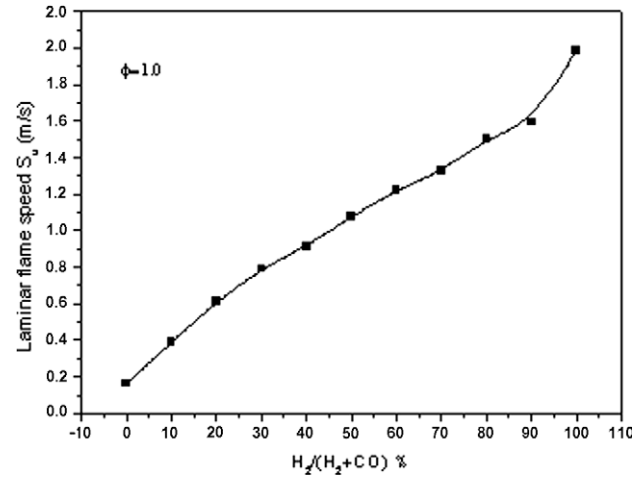


Fig. 7. Laminar flame speed versus H_2 fraction for H_2 /CO mixtures at $\phi = 1.0$.

predictions are a bit higher than the experimental results, this is mainly because the experimental condition is not adiabatic. Nevertheless, in general, the calculated speeds are in good agreement with the measured values.

4.3. Formulae to calculate laminar flame speed of H_2 /CO mixtures at different fractions and equivalence ratios

Basing on the experimental data, fitting curves of the laminar flame speeds for H_2 /air mixtures and CO/air mixtures are drawn in Figs. 9 and 10. The equations for calculating the laminar flame speed of H_2 /air and CO/air mixtures are given below:

$$S_{H_2} = 0.08925 + 1.59163\phi - 0.91917\phi^2 + 0.52964\phi^3 \quad (\phi = 0.7 - 2.1) \quad (4)$$

$$S_{CO} = 0.03276 + 0.18198\phi - 0.04156\phi^2 - 0.00791\phi^3 \quad (\phi = 0.7 - 2.1) \quad (5)$$

In order to calculate the laminar flame speed of H_2 /CO mixtures, we define $(S_x - S_{CO}) / (S_{H_2} - S_{CO})$ as the laminar flame speed increment. Here, S_x denotes the laminar flame speed at $x\%$ of H_2 fraction. Fig. 11 illustrates the increments of the laminar flame speed against the H_2 fraction of H_2 /CO mixtures. The correlation between the increment of the laminar flame speed and the H_2 fraction can be

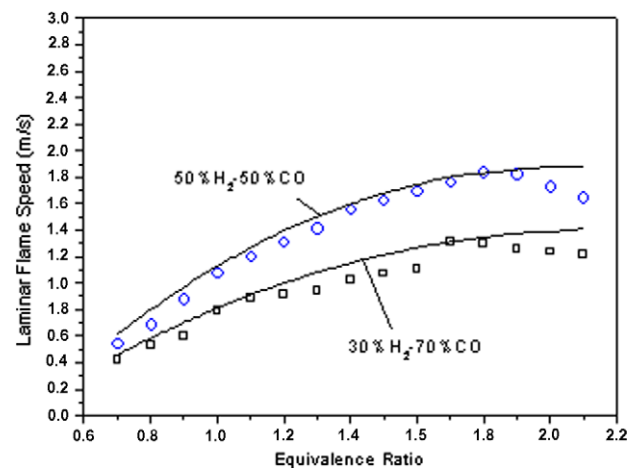


Fig. 8. Experimental (symbols) and computed (lines: Davis model [21]) H_2 /CO/air flame speeds at 1 atm and 20 °C.

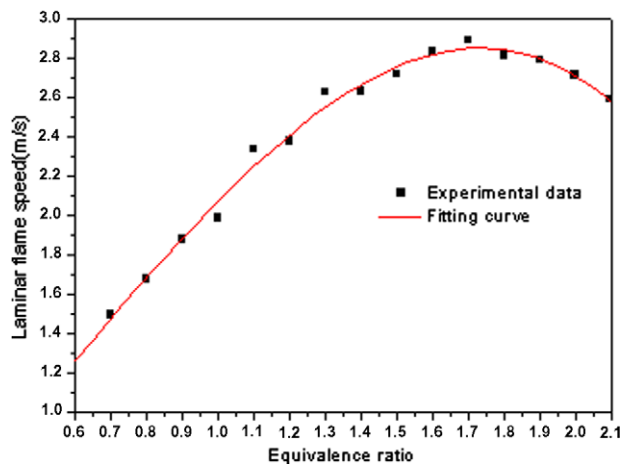


Fig. 9. Fitting curve of laminar flame speed versus equivalence ratio for H₂.

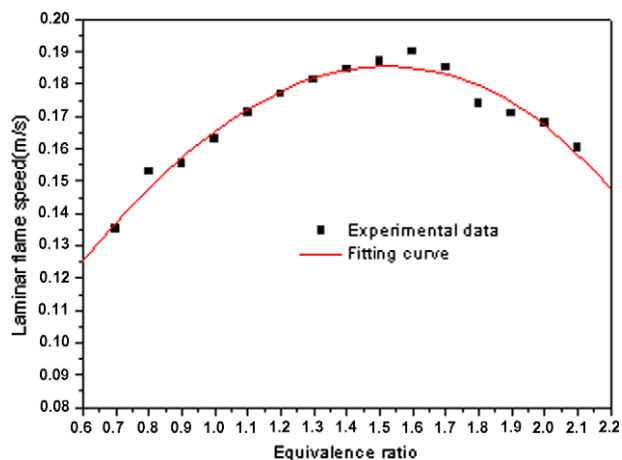


Fig. 10. Fitting curve of laminar flame speed versus equivalence ratio for CO.

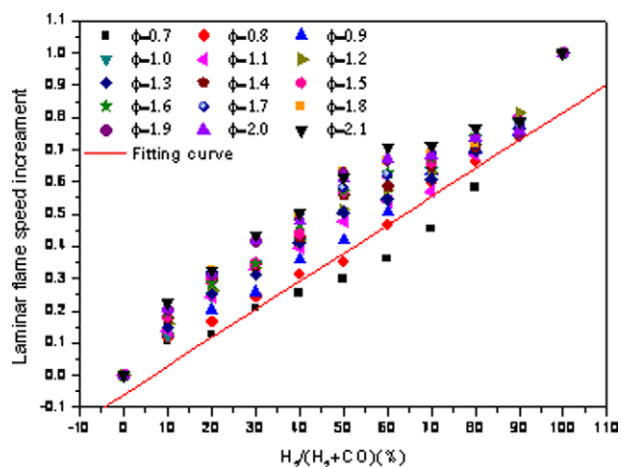


Fig. 11. Laminar flame speed increment versus H₂ fraction at different equivalence ratios.

fitted below, where S_{H_2} and S_{CO} are calculated by Eqs. (4) and (5) at a given equivalence ratio.

$$\frac{S_x - S_{CO}}{S_{H_2} - S_{CO}} = -0.05914 + 0.00875x \quad (\varphi = 0.7 - 2.1) \quad (6)$$

Using Eqs. (4)–(6), the laminar flame speed at different H₂ fractions (for a given equivalence ratio) can be easily calculated.

5. Conclusion

In this paper, the laminar flame speeds of H₂, CO and H₂/CO are studied in a Bunsen burner at normal temperature and pressure. The calculation of the laminar flame speeds is based on the reaction zone area instead of the unburned flame area. The good agreements among experimental results, and numerical simulations, and previous works have validated the proposed method. The influence of fuel composition and equivalence ratio on the laminar flame speed of the mixture is also analyzed. The main results are summarized as follows:

1. When the equivalence ratio equals to 1.7 and 1.6, the laminar flame speeds of H₂/air mixtures and CO/air mixtures reach their maximum values, 2.891 and 0.19 m/s, respectively.
2. The laminar flame speed of H₂/CO/air mixture increases with H₂ fraction. This increasing trend, as the H₂ volume fraction varies from 0% to 100%, can be divided to three regimes. The first one is a regime with rapid increment, the second one is linear variation, and the third one also rapidly increases.
3. Based on experimental data, fitting equations are proposed to effectively calculate the laminar flame speeds of H₂, CO and H₂/CO mixtures. Using these equations, the laminar flame speed at different H₂ fractions can be calculated for a given equivalence ratio.

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